

LESSON 12.6

REWRITING EXPRESSIONS WITH COMMON MONOMIAL FACTORS



LESSON INTRODUCTION

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

LEARNING TARGETS

- I can rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

WORKING TOWARD

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

ESSENTIAL QUESTIONS

- How do I factor two algebraic expressions that have a common monomial factor?
- How do I use the distributive property to create an equivalent algebraic expression?

LESSON NARRATIVE

In this lesson, students rewrite the sum of two algebraic expressions with a common monomial factor and check their work using the distributive property. This lesson builds upon what students practiced in Lesson 5, but removes the scaffolding of the area model that students used in prior lessons. Students will gain repeated and flexible practice with creating equivalent expressions by factoring expressions with a common monomial factor and using the distributive property to expand factored expressions to reveal an equivalent expression.

COMPONENT SUMMARY

12.6.1 WARM-UP (~5 MIN.)

Complete an equation and find the number of solutions

12.6.3 COLLABORATION (5-10 MIN.)

Analyze student work to determine if expressions are equivalent

12.6.5 YOUR TURN! (5-10 MIN.)

Factor or apply the distributive property to find equivalent expressions

12.6.2 GUIDED INSTRUCTION (10-15 MIN.)

Find and use the common monomial factor

12.6.4 ACTIVITY (20-25 MIN.)

Match equivalent expressions

12.6.6 WRAP-UP (~5 MIN.)

Reflect on the learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Monomial
- Greatest Common Factor

MATERIALS

- Matching Expression Cards

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe if two expressions are not written in the same order that they are not equivalent. Have students use the distributive property or area model to verify expressions are equivalent.

THINGS TO CONSIDER

- Students who lack mastery with the laws of exponents might struggle with this lesson. Consider having an Anchor Chart or reference sheet posted with the exponent laws, along with an example listed.
- If time is a concern, then consider assigning 12.6.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Stronger and Clearer Each Time

12.6.3 Collaboration provides an opportunity to use this language routine. For question 1, students first determine if they agree. All they share with their partner is if they agree (for questions 2 and 3 students would share who they agree with). After students state whether they agree, they explain what they know and how they know it is true. Students should be encouraged to use words, numbers, and diagrams as part of their explanation. Students should exchange explanations with their partners to read their partners' explanations. After reading their partner's explanation, students should take their papers back and add their "why" to their explanation. As part of explaining why, students are encouraged to use evidence and/or examples. Students should once again trade papers to read the added explanation from their partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

This lesson is an extension of the prior two lessons with a more complex set of problems. Throughout the lesson, students will adapt procedures to apply them to a new context and use feedback to improve efficiency when rewriting the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

DIFFERENTIATE INSTRUCTION

12.6.4 Activity offers student choice to create an additional equivalent expression. Consider a Stand-Pair-Share activity in which after students have completed the activity, they stand until the music being played by the teacher stops, pair with the closest classmate, then share one of their equivalent expressions. Students should compare equivalent expressions and if the expressions are different, they should verify both are equivalent to the original expression. Once the music starts playing again, students are to move around the room and repeat the same process, but with a different expression.

12.6.1 WARM-UP: MATCHING SOLUTIONS

TEACHER MOVES

12.6.1 Warm-Up is intended to spiral content from Unit 8 while also connecting to equivalent expressions using the distributive property. If students simplify the left side of the equation, the result is $12x-36+18$ which equals $12x-18$.

Question 1b asks specifically about which expression results in an equation with infinitely many solutions because this would mean both sides have equivalent expressions and $6(2x-3)$ is a factored form of $12x-18$.

Students do not have to think about the problem in this way as there are multiple pathways.



12.6.1 Warm-Up: Matching Solutions

1. Consider the unfinished equation $12(x - 3) + 18 = \underline{\hspace{2cm}}$.
 - a. Match the following expressions with the number of solutions the equation would have if that expression were on the right side of the equation.

Expression	Number of Solutions to the Equation
$6(2x - 3)$	one solution
$4(3x - 3)$	no solutions
$4(2x - 3)$	infinitely many solutions

- b. Explain how you determined which expression would result in an equation with infinitely many solutions.
Answers will vary. I selected the expression that is equivalent to the given expression on the left side of the equation.



12.6.2 Guided Instruction: Factoring Monomials

While the area model is helpful for visually representing the factors of an expression, it is not the only strategy.

Area models normally are used to represent multiplication but can be used in the reverse to find the factors of an expression. Similarly, the distributive property is normally used to multiply expressions but the reverse, division, can be used to find the factors multiplied.

1. Consider the expression $6x^5 + 2x^2$.
 - a. What is the greatest common factor of $6x^5$ and $2x^2$?
 $2x^2$

Recall the quotient of powers law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

- b. Divide both terms in the expression by the GCF determined in Part A.

$$\frac{6x^5}{2x^2} = 3x^3 \quad \frac{2x^2}{2x^2} = 1$$

- c. Rewrite the expression using the distributive property.

$$2x^2(3x^3 + 1)$$

- d. Verify using the distributive property that the expression you determined in Part C is equivalent to $6x^5 + 2x^2$.
 $2x^2(3x^3 + 1) = 6x^5 + 2x^2$

- e. Choose two different common factors to rewrite $6x^5 + 2x^2$.

$$2(3x^5 + x^2) \quad x(6x^4 + 2x)$$

- f. Confirm using the distributive property that the expressions are equivalent.
 $2(3x^5 + x^2) = 6x^5 + 2x^2$
 $x(6x^4 + 2x) = 6x^5 + 2x^2$

2. Consider the expression $9g^4h^5 - 12gh^3$.

- a. What is the GCF of the two terms in the expression?
 $3gh^3$
 - b. Rewrite $9g^4h^5 - 12gh^3$ using the GCF as one of the factors.
 $3gh^3(3g^3h^2 - 4)$
 - c. Verify using the distributive property that the expression you determined in Part B is equivalent to the original expression.
 $3gh^3(3g^3h^2 - 4) = 9g^4h^5 - 12gh^3$

12.6.2 GUIDED INSTRUCTION: FACTORING MONOMIALS

DIFFERENTIATE INSTRUCTION

Use observational data from the prior two lessons to determine how much guidance is needed to bring together key ideas in 12.6.2 Guided Instruction component. If necessary, consider modeling question 1a-c.

ENRICHMENT

Consider posing questions to solidify understanding as students work through this section.

- How do we find factors of whole numbers?
- How is the process different when we find factors of expressions?
- What is the difference between common factors and the Greatest Common Factor of an expression?
- Explain how to use the distributive property to verify your answers?

12.6.3 COLLABORATION: WHICH IS EQUIVALENT?

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 12.6.4 Activity. This could include highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.

DIFFERENTIATE INSTRUCTION

Stronger and Clearer Each Time

Consider using this language routine for questions 2 and 3. Students will first determine who they agree with. All they share with their partner is who they agree with. Students then write an explanation of what they know and how they know it is true. Students can use words, diagrams, etc. as part of their explanation. Students then trade explanations and read their partner's. After reading their partner's explanation, students should take their papers back and add "why" to their explanation. As part of explaining why, students are encouraged to use evidence and/or examples (or non-examples). Students should once again trade papers to read the added explanation from their partner.



12.6.3 Collaboration: Which is Equivalent?

Work with your partner to complete the following.

- Orly says the expressions $\frac{1}{2}x^3(8x + 10x^6)$ and $x^4(5x^5 + 4)$ are equivalent.
 - Do you agree with Orly?
Answers may vary. I agree with Orly.
 - Explain why or why not.
Answers will vary. I agree with Orly because both expressions simplify to $4x^4 + 5x^9$.
- Zeke and Ford each rewrote $3w^2x^2(3x^2 + 3w^6)$. Their expressions are shown.

Zeke's Expression	Ford's Expression
$9w^2x^4 + 3w^6$	$9w^2x^4 + 9w^8x^2$

- Who wrote an expression equivalent to $3w^2x^2(3x^2 + 3w^6)$?
 - ☐ Zeke
 - ☒ Ford
 - ☐ Both Zeke and Ford
- If one of the students wrote an expression that was not equivalent, write a note explaining how to avoid the same mistake in the future. If both students were correct, explain how they can both be correct.
Answers will vary. Zeke, remember to multiply the coefficient by each term in the parentheses.

3. Lucía and Faye each re-wrote $-21y^3 - 9y^4$ using a common monomial factor.

Lucía's Expression	Faye's Expression
$-3y^3(7 + 3y)$	$3y^3(7 - 3y)$

- a. Who wrote an expression equivalent to the original?
- Lucía
 - Faye
 - Both Lucía and Faye
- b. If one of the students wrote an expression that was not equivalent, write a note explaining how to avoid the same mistake in the future. If both students were correct, explain how they can both be correct.
- Answers will vary. Faye, remember that one of the factors must be negative in order to produce a negative product.*

12.6.3 COLLABORATION: WHICH IS EQUIVALENT? (CONTINUED)

DIFFERENTIATE INSTRUCTION

Encourage students to check each expression using the distributive property. Students who struggle to distribute may need a visual, such as an area model, to keep track of how to multiply the monomial by each term of the linear factor.

12.6.4 ACTIVITY: MATCHING EXPRESSIONS

QUESTIONING

Allowing students to create their own equivalent expressions is an opportunity to increase complexity during 12.6.4 Activity.



12.6.4 Activity: Matching Expressions

Six expressions are provided in the table.

Work with your partner to find two expressions from the card set that are equivalent to each given expression. Write the letter of each card in the columns labeled Card 1 and Card 2. Then, create an additional equivalent expression not included on either card in the last column.

Given Expression	Equivalent Card 1	Equivalent Card 2	Equivalent Expression
$14q^5r^2 - 28q^6r^4$	<i>L</i>	<i>G</i>	<i>Answers will vary.</i>
$9(-q^3r + 2q^2r^8)$	<i>E</i>	<i>J</i>	<i>Answers will vary.</i>
$\frac{2}{3}(9qr^3 + 18q^5r^2)$	<i>A</i>	<i>I</i>	<i>Answers will vary.</i>
$-q^7r^4 - 6q^4r^4$	<i>H</i>	<i>C</i>	<i>Answers will vary.</i>
$2(10q^4r^7 + 20q^2r^3)$	<i>B</i>	<i>D</i>	<i>Answers will vary.</i>
$-12q^3r + 15q^2r^5$	<i>K</i>	<i>F</i>	<i>Answers will vary.</i>



12.6.5 Your Turn!

1. Complete the table.

Expression	Factored Expression
$16t^6 + 20t^4$	$4t^4(4t^2 + 5)$
$-6k^4 + 15k^3$	$3k^3(-2k + 5)$
$21m^4n - 27mn^3$	$3mn(7m^3 - 9n^2)$

12.6.5 YOUR TURN!

TEACHER MOVES

If time is a concern, then consider assigning 12.6.5 Your Turn! for homework.

DIFFERENTIATE INSTRUCTION

Encourage students to use the area model for a visual representation. When students are expanding the factored expression, allow for students to use colored pencils or markers to show the distribution of the monomial to each term in the linear factor.



12.6.6 Wrap-Up: Reflection

1. Rate your understanding of today's learning target on the emoji scale.
Answers will vary.
 - ☐ I can rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.



12.6.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The rating scale provides an opportunity for students to self-assess their understanding. Consider having students explain their rating to better understand their thinking behind the rating they gave themselves. Consider using this assessment, along with 12.6.5 Your Turn! to inform remediation and small groups for future lessons.